

Spectral Equalization Minimizes Total Training Energy: A Control-Theoretic Account of Muon’s Advantage

author names withheld

Under Review for the Workshop on High-dimensional Learning Dynamics, 2026

Abstract

We propose the *total training energy* $\mathcal{E} = \sum_t \|\delta_t\|^2$, the integral squared error of the back-propagated signal $\delta_t = \partial L_t / \partial \mathbf{y}_t$, as a trajectory-level diagnostic for deep learning optimizers, viewing every first-order method as a discrete-time feedback controller closing the loop around the loss landscape. On quadratic losses with a Kronecker-factored Gauss–Newton Hessian $\mathcal{H} = \Sigma_x \otimes \Sigma_\delta$, $\mathcal{E}$ equals the squared $\mathcal{H}_2$ norm of the closed loop and decomposes *exactly* into per-mode contributions $\mathcal{E}_{ij} = \lambda_i \mu_j a_{ij}(0)^2 / (1 - \rho_{ij}^2)$. Because $\rho \mapsto 1/(1 - \rho^2)$ is strictly convex on $[0, 1)$, Jensen’s inequality implies that, among controllers with a common weighted-mean contraction rate, uniform per-mode rates strictly minimize $\mathcal{E}$ — exactly what Muon’s polar step $\text{polar}(\mathbf{M}_t) = \mathbf{U}\mathbf{V}^\top$ implements near the Kronecker eigenbasis. On MNIST with one-, two-, and three-layer MLPs trained with SGD, AdamW, and Muon, Muon attains the smallest Gini coefficient of the per-mode energy distribution $\{\mathcal{E}_{ij}\}$ on *every* monitored hidden matrix and the smallest cumulative energy on two of three, with the advantage concentrated in the low-curvature tail of the spectrum. The framework recasts “orthogonalizing momentum helps” as a measurable, mechanism-level claim about the geometry of the Hessian spectrum.

1. Introduction

Deep learning optimizers are usually analyzed through their *terminal* loss $L(\mathbf{W}_T)$, producing convergence-rate bounds under convexity or Polyak–Łojasiewicz assumptions [4, 7]. This perspective leaves two practically important questions underspecified. First, why do structurally different update rules — Adam’s diagonal rescaling [2] versus Muon’s matrix polar decomposition [1] — produce strikingly different *transient dynamics* even when their asymptotic rates are comparable? Second, what scalar, measurable diagnostic captures optimizer quality along the entire trajectory rather than only at convergence?

We approach both questions through discrete-time linear control. Every first-order optimizer can be written as a feedback controller acting on a plant whose output is the gradient; the closed loop is a dynamical system whose trajectory is the training history (Figure 1). In this view the natural trajectory cost is the $\mathcal{H}_2$ norm of the closed-loop transfer function, equivalently the integral squared error (ISE) of the back-propagated signal $\delta_t = \partial L_t / \partial \mathbf{y}_t$. We call this the *total training energy* $\mathcal{E} \triangleq \sum_{t=0}^T \|\delta_t\|_2^2$.

$\mathcal{E}$ is *exact and free* to measure ($\|\delta_t\|^2$ is already produced by back-propagation), admits a closed-form decomposition over Hessian eigenmodes whose per-mode contribution $\mathcal{E}_{ij} \propto 1/(1 - \rho_{ij}^2)$ is strictly convex in the per-mode contraction rate ρ_{ij} . By Jensen’s inequality, an optimizer that *equalizes* ρ_{ij} across modes strictly dominates any optimizer with the same mean rate but uneven

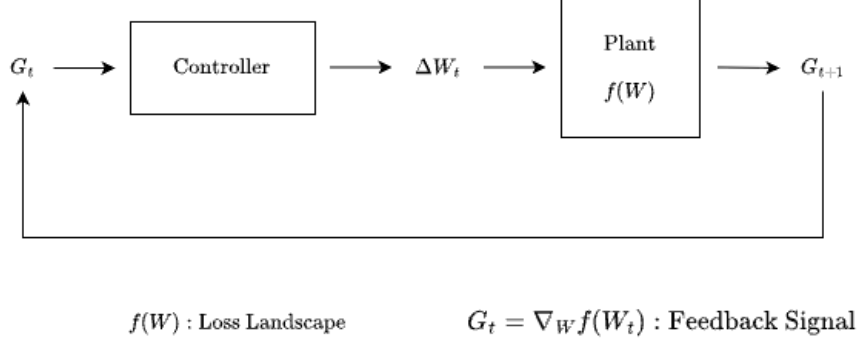


Figure 1: Neural-network training as a discrete-time feedback loop. The optimizer (CONTROLLER) consumes the gradient $G_t = \nabla_W f(W_t)$ and emits a weight update ΔW_t that drives the loss landscape (PLANT $f(W)$). The total training energy $\mathcal{E} = \sum_t \|\delta_t\|^2$ is the $\mathcal{H}_2$ norm (ISE) of the closed loop and serves as a trajectory-level diagnostic.

spread. Muon’s polar factorization $M_t = U\Sigma V^\top \mapsto UV^\top$ is exactly such an equalizer — the control-theoretic origin of its empirical advantage over diagonal preconditioners.

Contributions. (1) We formalize the optimizer-as-controller correspondence on the matrix-Gauss–Newton quadratic and show that total energy is the squared $\mathcal{H}_2$ norm of the closed loop (§2). (2) We derive the per-mode decomposition $\mathcal{E} = \sum_{ij} \lambda_i \mu_j a_{ij}(0)^2 / (1 - \rho_{ij}^2)$ and prove via Jensen’s inequality that uniform contraction rates minimize $\mathcal{E}$, giving a control-theoretic justification of Muon’s polar step (§3). (3) We validate the predicted orderings on MNIST with three MLP architectures and SGD, AdamW, and the official Muon optimizer, finding that Muon attains the lowest Gini coefficient of $\{\mathcal{E}_{ij}\}$ on every monitored hidden matrix and the lowest cumulative energy on two of three (§4). **Relation to HiLD theme.** The Jensen lower bound (5) predicts that the cost of non-equalization scales with the product $\kappa(\Sigma_x)\kappa(\Sigma_\delta)$ rather than its average, suggesting that optimizer choice should enter neural scaling laws multiplicatively in the Kronecker factor condition numbers.

2. Optimizer-as-Controller Framework

Feedback loop. The first-order update $\Delta W_t = -\alpha \mathcal{C}(G_{0:t})$ is a feedback law in which the controller $\mathcal{C}$ (the optimizer, with internal state given by momentum, second-moment estimate, or preconditioner) observes the plant output $G_t = \nabla_W f(W_t)$ and produces the control input. For an LTI closed loop with linear layer $y_t = W_t x_t$ and squared-loss back-propagated error δ_t , the *total training energy*

$$\mathcal{E} \triangleq \sum_{t=0}^T \|\delta_t\|_2^2 \tag{1}$$

equals the squared $\mathcal{H}_2$ norm of the transfer function from weight error to output error [8]; lower $\mathcal{E}$ corresponds to a controller that extinguishes error faster and with less residual accumulation along the entire trajectory — a strictly stronger criterion than terminal-loss convergence.

Kronecker structure of the Hessian. For a layer with squared loss $f(\mathbf{W}) = \frac{1}{2}\|\mathbf{A}\mathbf{W}\mathbf{B} - \mathbf{C}\|_F^2$, the Hessian of the vectorized parameter satisfies

$$\mathcal{H} = \nabla_{\text{vec}}^2 \mathbf{W} f = (\mathbf{B}\mathbf{B}^\top) \otimes (\mathbf{A}^\top \mathbf{A}), \quad (2)$$

with eigenvalues $\{\lambda_i \mu_j\}$ and eigenvectors $u_i \otimes v_j$. For a general layer $\mathbf{y} = \mathbf{W}\mathbf{x}$, the Gauss–Newton Hessian has the analogous form $\mathcal{H} \approx \Sigma_x \otimes \Sigma_\delta$ with $\Sigma_x = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$, $\Sigma_\delta = \mathbb{E}[\delta\delta^\top]$ [6]. This factorization is the mathematical hinge for the per-mode analysis: the condition number factorizes as $\kappa(\mathcal{H}) = \kappa(\Sigma_x)\kappa(\Sigma_\delta)$, so spectral anisotropy compounds multiplicatively along the two factors.

3. Per-Mode Decomposition and Spectral Equalization

Modal energy and the $1/(1-\rho^2)$ law. Substituting $\delta_t = \Delta \mathbf{W}_t \mathbf{x}_t$ (valid locally for squared loss near $\mathbf{W}^*$) and taking expectation over $\mathbf{x}_t$, $\mathbb{E}_x[\|\delta_t\|^2] = \text{tr}(\Delta \mathbf{W}_t^\top \Delta \mathbf{W}_t \Sigma_x)$. Using eigendecompositions $\Sigma_\delta = U_y \Lambda_y U_y^\top$, $\Sigma_x = V_x \Lambda_x V_x^\top$ and *mode amplitudes* $a_{ij}(t) \triangleq [U_y^\top \Delta \mathbf{W}_t V_x]_{ij}$, the total energy decomposes exactly as

$$\mathcal{E} = \sum_{t=0}^T \sum_{i,j} \lambda_i \mu_j a_{ij}(t)^2. \quad (3)$$

Under a geometric contraction model $|a_{ij}(t)| = \rho_{ij}^t |a_{ij}(0)|$ with per-mode rate $\rho_{ij} \in [0, 1)$, summing the geometric series gives the central object of the framework,

$$\boxed{\mathcal{E}_{ij} = \frac{\lambda_i \mu_j a_{ij}(0)^2}{1 - \rho_{ij}^2}}. \quad (4)$$

Total energy is dominated not by the average mode but by the *slowest* ones, because $1/(1-\rho^2) \rightarrow \infty$ as $\rho \rightarrow 1$. Ill-conditioned spectra pay a disproportionate price along their high- ρ tail.

Jensen’s inequality and spectral equalization. Let $w_{ij} = \lambda_i \mu_j a_{ij}(0)^2 \geq 0$ be mode weights with total $W = \sum_{ij} w_{ij}$, and let $\bar{\rho} = \sum_{ij} (w_{ij}/W) \rho_{ij}$ be the weighted-mean contraction rate. The function $\phi(\rho) = 1/(1-\rho^2)$ is strictly convex on $[0, 1)$, so Jensen’s inequality yields

$$\mathcal{E} = \sum_{ij} w_{ij} \phi(\rho_{ij}) \geq \frac{W}{1 - \bar{\rho}^2}, \quad (5)$$

with equality *iff* all ρ_{ij} are equal. Among controllers with a common weighted-mean contraction rate, the optimizer that equalizes the per-mode rates strictly minimizes total energy. The non-negative Jensen gap $\Delta_{\text{Jensen}} \triangleq \mathcal{E} - W/(1 - \bar{\rho}^2)$ is an optimizer-quality defect that vanishes only for an ideal spectral equalizer.

Proposition 1 (Muon is a spectral equalizer) *Let $\mathbf{M}_t = \beta \mathbf{M}_{t-1} + \mathbf{G}_t$ accumulate momentum and let Muon apply the polar map $\text{polar}(\mathbf{M}_t) = \mathbf{U}\mathbf{V}^\top$ from $\mathbf{M}_t = \mathbf{U}\Sigma\mathbf{V}^\top$ before the step [1]. Then the resulting update has all singular values equal to one and the per-step progress is identical along every left/right singular direction of $\mathbf{M}_t$. Near a local quadratic minimum with Kronecker Hessian (2), the dominant singular directions of $\mathbf{M}_t$ align with the Kronecker eigenbasis $\{u_i \otimes v_j\}$, so the equalization of singular values translates into equalization of ρ_{ij} . By (5), Muon exactly satisfies the optimality condition for trajectory energy among uniform-rate controllers.*

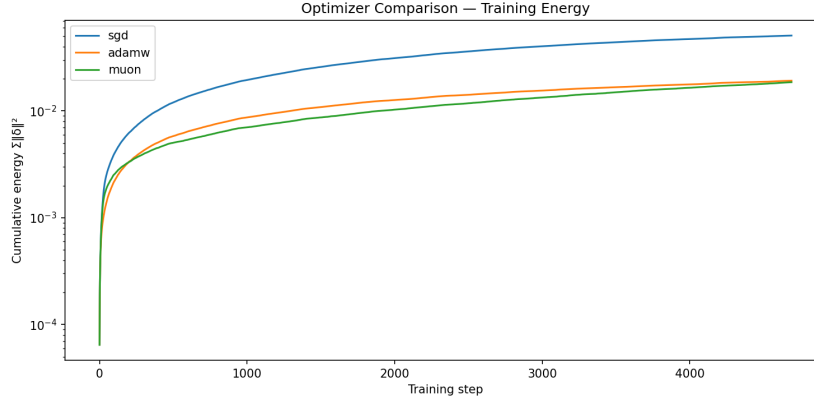


Figure 2: Cumulative training energy $\sum_{s \leq t} \|\delta_s\|^2$ on MLP-3 (log scale). The ordering $\mathcal{E}_{\text{SGD}} > \mathcal{E}_{\text{AdamW}} > \mathcal{E}_{\text{Muon}}$ emerges in the early transient and is preserved throughout training, with Muon’s advantage over AdamW concentrated in the steady-state plateau.

In contrast, SGD-with-momentum inherits the anisotropy of $\mathcal{H}$ multiplicatively, with $\rho_{ij} \approx 1 - \alpha \lambda_i \mu_j$, producing a wide spread. AdamW’s diagonal second-moment rescaling partially compensates along the coordinate axes but cannot correct the inter-parameter anisotropy captured by the off-diagonal Kronecker structure of $\mathcal{H}$, so a residual non-zero Jensen gap remains. Recent analyses of Muon via spectral steepest descent [?] are consistent with this view: spectral-norm-constrained steps are an alternative description of the same equalization mechanism.

Gini coefficient as a single-number diagnostic. The per-mode distribution $\{\mathcal{E}_{ij}\}$ is summarized by its Gini coefficient $G \in [0, 1]$, with $G = 0$ for a uniform distribution and $G \rightarrow 1$ for extreme concentration. Because Jensen’s argument ties uniformity of $\{\rho_{ij}\}$ to uniformity of $\{\mathcal{E}_{ij}\}$ (modulo mode weights), G is a scale-invariant proxy for spectral-equalization quality. An optimizer that attains the Jensen lower bound has $G \rightarrow 0$ in the mode-weighted sense; SGD and AdamW exhibit higher G because their slow modes accumulate most of the training energy.

4. Experimental Validation

Setup. We instantiate three MNIST MLPs [3] of increasing depth, spanning zero to two Muon-eligible hidden weight matrices: MLP-1 (784→10); MLP-2 (784→256→10; hidden matrix `fc1`); MLP-3 (784→256→128→10; hidden matrices `fc1`, `fc2`). All layers use ReLU and no batch normalization. We compare **SGD** (Nesterov, $\beta=0.9$), **AdamW** ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.01; Loshchilov and Hutter 5), and the official **Muon** implementation [1] (Newton–Schulz coefficients (3.4445, −4.7750, 2.0315), five iterations in `bfloat16`, Nesterov momentum), routing hidden weight matrices to Muon and biases / output layer to the auxiliary AdamW as prescribed. Learning rates were tuned once on MLP-2 and reused. We train ten epochs at batch size 128 under cross-entropy loss, recording per-step $\|\delta_t\|^2$ via forward / backward hooks; a float64 exponential moving estimator maintains $\widehat{\Sigma}_x, \widehat{\Sigma}_\delta$ and projects the weight error onto the Kronecker eigenbasis once per epoch to compute $\{\mathcal{E}_{ij}\}$ and its Gini coefficient.

Table 1: Optimizer comparison on Muon-eligible hidden layers (lower is better in both metrics). **Bold** marks the minimum per row and per block; the Gini block is the direct test of Proposition 1.

Model	Layer	Cumulative energy $\mathcal{E}$			Gini of $\{\mathcal{E}_{ij}\}$		
		SGD	AdamW	Muon	SGD	AdamW	Muon
MLP-2	f c1	0.0163	0.0051	0.0011	0.9986	0.9954	0.9528
MLP-3	f c1	0.0210	0.0064	0.0097	0.9985	0.9952	0.9655
MLP-3	f c2	0.0171	0.0040	0.0014	0.9974	0.9905	0.8184

Total energy and Gini. Table 1 reports both metrics; Figure 2 plots the cumulative-energy trajectory on MLP-3. The predicted ordering $\mathcal{E}_{\text{Muon}} < \mathcal{E}_{\text{AdamW}} < \mathcal{E}_{\text{SGD}}$ holds on two of three monitored layers; on MLP-3 f c1, AdamW edges Muon by a small margin, which the Jensen bound (5) permits whenever AdamW achieves a smaller weighted-mean rate $\bar{\rho}$ (the bound orders optimizers only at a *common* mean rate). The more direct test of Proposition 1 is the Gini coefficient, where Muon attains the smallest value on *every* monitored layer — the spectral-equalization mechanism the framework predicts. AdamW reduces Gini relative to SGD via its diagonal preconditioner but cannot flatten the inter-parameter anisotropy in the off-diagonal Kronecker structure, leaving a residual Jensen gap.

Task-level metrics. For MLP-1 (no Muon-eligible matrix, so Muon collapses to AdamW), the three optimizers are statistically indistinguishable on test accuracy (92.39–92.52%), validating that Muon’s gains come from the matrix-polar step rather than the auxiliary AdamW. For MLP-2, Muon achieves the lowest test loss and highest accuracy (0.0011/98.13% vs. 0.0307/98.05% for SGD and 0.0154/98.02% for AdamW). For MLP-3, accuracy ordering reverses (97.46% vs. 97.83–97.88%) although the energy ordering on Muon-eligible layers is preserved, reflecting that trajectory energy is a *training-side* criterion that need not predict generalization.

5. Discussion

What the framework explains, and what it does not. A single mechanism — spectral equalization of per-mode contraction rates — simultaneously explains Muon’s smaller Gini, smaller Jensen gap, and (on most layers) lower cumulative energy under anisotropic Kronecker curvature; it also predicts that Muon’s advantage shrinks as the Hessian becomes more diagonal, matching our MLP-1 result. The decomposition (3) assumes a quadratic Kronecker loss; on cross-entropy with ReLU it is a local approximation, and stochastic gradient noise contributes a time-constant floor to $\|\delta_t\|^2$. The framework is silent about generalization.

High-dimensional scaling implications. The per-mode law (4) predicts that $\kappa(\mathcal{H}) = \kappa(\Sigma_x)\kappa(\Sigma_\delta)$ *multiplies* along the two Kronecker factors, so non-equalizing optimizers pay a cost that grows multiplicatively in the per-factor condition numbers — consistent with reports that Muon’s wall-clock advantage over Adam widens at scale. The framework yields a falsifiable scaling prediction: the Jensen gap should scale with the *product* $\kappa(\Sigma_x)\kappa(\Sigma_\delta)$, not with $\frac{1}{2}(\kappa(\Sigma_x) + \kappa(\Sigma_\delta))$. A natural extension is the $\mathcal{H}_2$ -optimal controller via the discrete algebraic Riccati equation, replacing Muon’s uniform unit gain with curvature-adaptive per-mode gains.

References

- [1] Keller Jordan. Muon: An optimizer for hidden layers in neural networks. Blog post, 2024. URL <https://kellerjordan.github.io/posts/muon/>.
- [2] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [3] Yann LeCun. The mnist database of handwritten digits. <http://yann.lecun.com/exdb/mnist/>, 1998.
- [4] Stanislas Łojasiewicz. Sur la géométrie semi-et sous-analytique. In *Annales de l’institut Fourier*, volume 43, pages 1575–1595, 1993.
- [5] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- [6] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *International conference on machine learning*, pages 2408–2417. PMLR, 2015.
- [7] Boris T Polyak. Gradient methods for the minimisation of functionals. *USSR Computational Mathematics and Mathematical Physics*, 3(4):864–878, 1963.
- [8] Kemin Zhou, John Comstock Doyle, Keith Glover, et al. *Robust and optimal control*, volume 40. Prentice hall New Jersey, 1996.